BULK FERMENTATION

When making some of our doughs, such as our Focaccia Dough (see page 148) and New York Square Pizza Dough (see page 152), the mixing stage is followed by bulk fermentation, also known as first proof or first fermentation. You are simply fermenting your dough at room temperature in one piece before dividing it into smaller portions. The duration of bulk fermentation is most often determined by three factors: how long the dough was mixed, the consistency of the dough, and the type of yeast (commercial, levain, or a combination) used. The bulk fermentation stage is standard in bread baking but far less common among pizzaioli, who tend to go straight from mixing their dough to balling it for the fermentation stage.

We include the bulk fermentation stage for our Neapolitan dough and for our bread-like pizzas. Neapolitan dough is not traditionally bulk fermented, but after extensive testing for the best method to proof our master Neapolitan dough (see page 124), we found it gave us the best results.

The amount of mixing time is generally inversely proportional to the duration of bulk fermentation. That is, the longer the mix (the more the gluten is developed), the shorter the bulk ferment; the shorter the mix, the longer the bulk ferment. The dough is developed during bulk fermentation through a series of folds (see page 50). The duration matters because it allows time for enzymatic activity and hydration to develop gluten; this happens when the gluten strands organize and through the acidification that occurs during a long fermentation. Acidity helps strengthen gluten strands, whether naturally produced by the yeast (commercial or wild) or added in the form of ascorbic acid during mixing (see page 1:324).

Note that the majority of our doughs are *not* bulk fermented. Instead, they're mixed to full gluten development and then go through what's called bench rest. This 15- to 20-minute rest period after mixing allows the dough to relax before dividing and preshaping. Our master Neapolitan dough is an outlier in that it's mixed to full gluten development *and* allowed to bulk ferment at room temperature in a tub for 20–24 hours.

We bulk ferment any doughs that are mixed to medium gluten development, including those that have an inclusion. We incorporate inclusions at the

Folding during bulk fermentation redistributes the yeast and carbon dioxide in the dough as well as evens out the temperature of the dough.

medium gluten development stage (see page 30) because if the gluten is fully developed, the dough will be too strong to easily take in the inclusion, which might cause tearing. Adding the inclusions at this halfway point is much gentler on the dough. We finish developing the dough during bulk fermentation through a series of gentle four-edge folds (see page 51).

THE BENEFITS OF BULK FERMENTING

Bulk fermentation allows the dough to rest after mixing, which gives the enzymes enough time to act on the components of the dough. Proteases (protein-eating enzymes) cut gluten strands into smaller strands by breaking down the peptide bonds between amino acid chains in the proteins. The act of mixing and folding strengthens these strands even as they get smaller and smaller because the latticework of the gluten network becomes more interconnected. The gluten will eventually form a network of chains and strands that become the structure of the dough. Protease is found in very small amounts; an excess of it would cut gluten strands too much and have the opposite effect. If the dough is allowed to autolyse (see page 32) or if a preferment is used, proteases will have enough time to do their work. Bulk fermentation gives these enzymes even more time. It strengthens the dough slowly and gradually without adding excessive oxygen as with long mixing times.

FACTORS THAT AFFECT BULK FERMENTATION

Multiple factors influence the time the dough spends bulk fermenting at room temperature, including dough hydration, mixing method, temperature and humidity, and yeast type. High-hydration doughs require a longer bulk fermentation time and additional folds; this allows time for the gluten to form a firm structure in what is a very slack dough. A dough mixed by hand will require a longer bulk fermentation time and more folds to strengthen its gluten structure than a dough mixed by machine to medium gluten development. Doughs that are machine-mixed to low gluten development will also require more time to bulk ferment and an increased number of folds.

The warmer the room, the faster the dough will ferment. For most doughs, try to maintain a cool environment (approximately 21°C / 70°F). If you're unsure, it's always best to err on the side of caution and go with a cooler temperature. Maintaining a consistent temperature is also an important consideration, as temperature fluctuations during bulk fermentation can impact the duration. Some kitchens have access to a proofer (see page 74), which allow you to bulk ferment your dough at a specific, consistent temperature. If you have a temperature-controlled pizzeria, however, you might not need a proofer-retarder.

A dough that contains commercial yeast will ferment much faster than a dough that contains only levain. In theory, a dough made with commercial yeast should generally be bulk fermented for a relatively short time compared to a levain-based dough, which benefits from a long bulk ferment and multiple folds. We say "in theory" because you can simply reduce the amount of commercial yeast in your dough and extend bulk fermentation time. This is key for high-hydration doughs that are mixed by hand.

Neapolitan Pizza Dough (see page 124)

STRATEGIES FOR BULK FERMENTATION

Type of mix	Length of bulk fermentation	Number of four-edge folds (unless noted in the recipe)	Recommended doughs	Doughs not recommended
medium gluten development	2–2½ h	4–6	focaccia dough New York square pizza dough high-hydration al taglio pizza dough all doughs with inclusions	thin-crust pizza dough Brazilian thin-crust pizza dough deep-dish pizza dough Neapolitan pizza dough New York pizza dough artisan pizza dough Detroit-style pizza dough
full gluten development	the dough is bench rested instead for 15–20 min before dividing	0	thin-crust pizza dough Brazilian thin-crust pizza dough deep-dish pizza dough Neapolitan pizza dough New York pizza dough artisan pizza dough Detroit-style pizza dough	New York square pizza dough high-hydration al taglio pizza dough foccacia dough